

Asynchronous Collaboration Systems for Evolving Information

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ABSTRACT

I aim to understand how individuals may be motivated non-financially to collaborate in maintaining evolving information asynchronously. I have developed two systems, Drafty and Sketchy, that provide evidence of achieving this goal at differing timescales. In Drafty, unpaid contributors voluntarily update a tabular dataset of Computer Science professor profiles at a multi-year timescale. In Sketchy, university students simultaneously sketch and co-inspire each other during voluntary UI/UX sketching activities at a sub-hour timescale. I seek to integrate lessons from both systems to develop a theoretical framework that uses a feedback cycle to provide users with practical random examples from evolving information to motivate new contributions. I will further develop this theoretical framework by extending Drafty to generate insights to automatically attract visitors and create the Virtual Readability Lab (VRL) to help users improve their reading performance through personalized readability tests and deepen our understanding of information design human performance. Both systems will give users the agency to seek randomized insights within a naturalistic setting to motivate continuous contributions to each system's evolving information.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**; **Computer supported cooperative work**; *Asynchronous editors*; • **Information systems** → **Crowdsourcing**; *Data cleaning*; Incomplete data.

KEYWORDS

evolving information, asynchronous collaboration, crowdsourcing, unpaid contributors

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1 RESEARCH SITUATION

In my research, I focus on developing and deploying systems to maintain evolving sets of information. The critical question is how

to use contributors' interest in the data to motivate further contributions. I have developed two web applications to study this question at different timescales. Drafty, a smarter Wiki for maintaining tabular data in the familiar form of a freely editable spreadsheet [34, 37]. Drafty hosts a tabular dataset of computer science professor profiles maintained through unpaid contributions in the wild. The second is Sketchy, which allows instructors to bring any sketching task to a UI/UX classroom where students can simultaneously create and update their sketches while co-inspiring each other through the "Peek" feature [33]. Each system demonstrates how users can be motivated to make small unpaid contributions to maintain and grow evolving information.

I am a fifth-year Ph.D. candidate in the Department of Computer Science of Brown University advised by Jeff Huang. I have completed the required coursework and passed the necessary comprehensive exams consisting of research and programming comps. I have also reached the residency requirement at my university. My projected thesis proposal date is Summer 2022, and my projected graduation date is Fall 2022. I began my graduate career at Brown University as a part-time Master's student in 2014. I started the Ph.D. program in Fall 2017 as a full-time student. From Spring 2018 and on, I have been a part-time Ph.D. student working full-time as the Lead Systems Programmer for Brown University's Computer Science Department. During the summer of 2019, I completed a research internship at the Adobe Creative Intelligence Lab under the supervision of Zoya Bylinskii.

2 CONTEXT AND MOTIVATION

The data around us is dying [12, 25]. Not all data is static, some data evolves over time due to temporal and external factors [34]. For example, University closures and mask requirements during the pandemic can change rapidly [1]. Whereas the course a university offers in Fall 2019 will not change. While we have studied how to quickly and cheaply collect accurate data, the life cycle of this data relies on maintenance to ensure its accuracy over time [34]. There are numerous collective Peer Production and Citizen Science efforts to build these types of evolving datasets, such as FLOSS, Wikis, and Knowledge Graphs. These datasets, powered by human-collaboration, have life cycles where data is collected, matures, and eventually dies [13]. The coming death of specific datasets is well documented on popular platforms like Wikipedia [12, 13, 19, 27]. Wikipedia and WikiData, for example, rely on a few users making the majority edits per article or topic [13, 16, 25]. When editors leave, data can rapidly become out-of-date. There is even a Wikipedia project [39] dedicated to absent or retired editors with more than 1000 edits. These former editors often possessed valuable domain expertise lacking among paid crowdworkers [27, 34]. Extending the life of data requires perpetual human attention. External temporal factors cause data to decay, thus, destroying its utility [34]. To overcome this, past work has employed

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paid-crowdworkers in crowd-powered systems [4] or allowed users to compare their performance in online tests with others to motivate voluntary participation [26].

Based on my experience with three diverse systems, I have begun extracting general principles for promoting data maintenance of evolving information through cyclical feedback mechanisms between systems and their users [30, 32, 33, 35, 37]. I intend to develop these into a framework for developing asynchronous collaboration systems for evolving information. I have two research goals to extend this theoretical framework. First, I will extend Drafty to continuously attract and engage users by automatically providing random insights, or DataBaits, after each unpaid contribution. The second is to develop the Virtual Readability Lab building on my suite of remote readability tests, where unpaid users are provided randomly designed text within personalized readability tests to understand how to individuate text.

3 PRELIMINARY WORK

3.1 Drafty: Maintaining Tabular Data

My first effort to maintain evolving information began by developing Drafty in 2015. Drafty hosts a publicly editable spreadsheet of Computer Science faculty profiles [37]. While paid crowdworkers originally collected this dataset [23, 34], the initial efforts to maintain it within our research lab proved unsuccessful due to time constraints and the growing complexity and size of the information. Over time, we conducted two case studies to observe paid crowdworkers and unpaid contributors' ability to improve the accuracy of the data. Our results show evidence that unpaid contributors edited frequently and were more accurate, especially for information requiring domain-specific knowledge [34].

3.1.1 Description and Lessons Learned Developing the Theoretical Framework.

Unpaid Users. Attracting interested visitors to make unpaid contributions is necessary to extend the information's maintenance phase. When Drafty was created, we made posts across various CS forums and websites (Reddit CompSci, Hacker News, and TheGradCafe) to inform and attract an initial user base interested in Computer Science. All subsequent visitors to Drafty was generated through organic search traffic or other means we did not control.

Evolving Information. Drafty hosts a publicly editable tabular dataset over 6,000 academic profiles of Computer Science tenure track faculty from the United States and Canada [37]. Each academic profile is a row in a spreadsheet. The columns corresponds to a professor's affiliated university, the year they joined as faculty, subfield area of expertise, and where they received their Bachelors, and Ph.D. degrees. This information of Computer Science faculty is valuable to the community, and powers one ranking within the meta-ranker CS Open Rankings [31].

External Temporal Factors. The information in a Computer Science professor's academic profile that is not static is their full name, the university they are employed at, the year they joined their current university, and their subfield area of expertise (i.e., primary research area). The external factors that cause the data to change include a professor moving universities, a new hire at a university, a

professor retiring or leaving to go to an industry position, or changing their subfield area of expertise. Thus, these external factors cause the data to change annually due to tenure appointments or hiring season, or slower if a professor changes their primary research field. We have found it unlikely that a tenure track faculty would receive a new Bachelors's or Doctorate.

Randomized Examples to Motivate Contributions. Our initial version of Drafty would solicit users to review and fix data matching their interests [37]. Drafty derives a user's interests from their interactions (i.e., clicks, searches, and edits). Drafty first selects rows the user is interested in and then randomly a row and value to propose a value to fix. Selecting values at random ensures a user is not asked to fix the same row of data multiple times.

The Reward to Motivate Engagement. The initial intention of Drafty was that the community's interest in the data was sufficient motivation to maintain it [34, 37]. While asking users to fix data matching their interests provided accurate edits, this reward did not provide enough immediate utility to the user to motivate frequent edits.

3.2 Sketchy and the Peek Feature

Students during a sketching task can benefit from seeing related examples; however, these can be difficult to collect and maintain [15, 29, 33]. We also observed people physically peeking at their neighbor's sketches in real-time to gain ideas [33]. To create this behavior in a digital environment, we developed a web-based drawing application, Sketchy, where users sketch in virtual rooms and use the Peek functionality to gain ideas from their peers' sketches in real-time [33]. Peek allows students to co-create the inspirational stimuli needed to conduct a helpful sketching activity, thus eliminating the need for a professor to collect and maintain prior sketching examples.

3.2.1 Description and Lessons Learned Developing the Theoretical Framework.

Unpaid Users. Sketchy's users consist of university students from a user experience and user design class. The students voluntarily participated in sketching activities during a break at a lab and right after class. Students did not receive a grade, and a few students were randomly awarded a gift card to a local restaurant. Thus the participation was voluntary, and students could leave it anytime. Our pilot studies used to reward the students whose sketches were voted most inspirational by other students with gift cards. However, we noticed this gamification produced adverse behaviors from specific students trained to game the system. Thus the most beneficial behaviors are exhibited when student participation was entirely voluntary with a small motivator in the form of a gift card raffle.

Evolving Information. The sketch produced by a user in Sketchy is the information. Sketchy automatically stores and synchronizes everyone's sketches as series of XY points and color per stroke. This enables one user's sketch to be seen by another user on-demand in real-time, as a form of digital peeking. Prior work shows individuals can benefit from seeing examples of other sketches during sketching activities [15, 29, 33]. It is difficult for an instructor to collect enough related sketch examples before running a sketching activity. Through the Peek feature, Sketchy provides a mechanism to quickly collect and share sketches as inspirational stimuli to users in real-time.

External Temporal Factors. Sketchy was developed to run hybrid remote and in-person UI/UX sketching tasks in classrooms. Over time relevant examples for a given sketching task will change due to various factors. For example, a sketching task from 20 years ago might not include designing a wireframe for a responsive web page on desktops, tablets, and phones. Over time different users' understanding of icons and their meaning will gradually shift for different age groups [2], thus necessitating a different set of inspirational examples to design icons. Overcoming these factors is complex, and Sketchy assumes that the current group of users can co-inspire their peers during the sketching process.

Randomized Examples to Motivate Contributions. In Sketchy, users sketch on their device during voluntary sketching activities and can freely peek (i.e., view another person's sketch) at their peers' in-progress sketches. These sketches are selected randomly and will not show users the same sketch twice unless they view all other sketches. Our pilot studies explored using Machine Learning models to recommend the most inspirational sketch. However, this choice frustrated users because they wanted to see a variety of sketches. The development of the digital Peek feature was informed by our qualitative observations of people physically peeking during in-person sketching tasks to gain inspiration. The Peek feature enables students to rapidly co-inspire and creates sketches in real-time. This instant co-creation of inspirational stimuli allows teachers to theoretically run any sketching task without the need to collect previous sketching examples to assist students' individual creative processes.

The Reward to Motivate Engagement. Sketchy's users are students in UI/UX classes with an intrinsic interest and motivation to develop their design skills. Peek provides the reward mechanism to increase users' overall creativity and satisfaction with their final sketch [33]. Peek delivers a method for users to see another sketch develop in real-time and copy other users' ideas. This allows users to see if their ideas are in other sketches and compare their sketches' quality in real-time. Prior research shows that this comparison method can motivate user engagement [26].

4 THESIS STATEMENT

Motivation

The life cycle of information follows the phases of growth, maturity, and decline. While paying crowdworkers can enable the growth phase, humans must continuously sustain the maturity phase as information evolves.

Thesis Statement

Providing users with individual random examples they can use immediately motivates them to interact continuously with and contribute new information. This thesis develops these ideas through a series of asynchronous collaboration systems where users' unpaid contributions extend the maturity phase of evolving information.

5 PROPOSED WORK

The proposed work will apply the theoretical concepts, the *who*, *what*, *when*, *where*, and *why* of the theoretical framework (see Section §5.1), for asynchronous collaboration systems by extending Drafty [34, 37] and developing the Virtual Readability Lab using my remote readability tests [3, 30, 32, 35, 36, 38].

5.1 Theoretical Concepts for Asynchronous Collaboration Systems for Evolving Information

Unpaid Users (who): The users will be interested in the system's information, proving beneficial to their daily lives or activities, thus attracting users to interact with the system voluntarily. These users are considered unpaid contributors and will likely have the domain expertise required to maintain domain-specific knowledge [34] that paid crowdworkers often lack [7, 10]. While paying crowdworkers can help collect information to bootstrap a system, paying them continuously to maintain information is not feasible [34]. This might be due to varying levels of motivation [22] and effort [5], malicious crowdworkers [11], and insufficient domain expertise [28]. A system can decrease vandalism by providing users with accurate information and immediate insights, thus creating a mutually beneficial feedback cycle between the user and the system [34, 37].

Evolving Information (what): The system contains information that is interesting and valuable to users, evolves due to external factors, and is difficult to collect and maintain by a single person [37]. Not all data is static; to maintain its accuracy and benefit over time, it is necessary for groups of people to collectively engage with the information to improve its quality [34]. Static data does not require continuous review, and it can be collected once and then immediately verified to improve its accuracy [34]. The information itself should be important to its community of unpaid contributors, thus providing benefits outside the system's confines. This will motivate users to visit and interact with the system to maintain its information.

External Temporal Factors (when): External factors outside the system cause information to become out-of-date. These factors are temporal, as the accuracy of some data is long-lived while others can change rapidly. Understanding what factors cause data to change is vital to understanding when users might change it and inform integrating new features and reward mechanisms [34]. Developing rewards that focus on evolving information instead of static will benefit the system. Understanding these external factors and designing a system and reward mechanisms to react to them is one method to maintain information [33, 34].

Randomized Examples to Motivate Contributions (where): A user, not the system, can initiate a random example from the system during their natural information seeking behaviors [37]. The random example can contain a recommendation or insight derived from the system's evolving information to help the user, and it should be useful within and outside the system. If the number of possible random examples is high, they can be filtered using an unpaid contributor's interests or prior behaviors [37]. A random example provided in the moment can influence a user's next action to help them continuously engage and provide accurate and useful information for other users [6, 33]. The random example should appeal to their intrinsic and altruistic motivations [33–35]. Random examples provide a means to implicitly match subjective information requiring domain-specific knowledge, with the domain expertise and interests of unpaid contributors to offer a diverse set of interpretations [34, 35].

The Reward to Motivate Engagement (why): The system should reward the user for completing the randomized examples to motivate the user to visit, engage, and contribute new information. Paying

crowdworkers is a common extrinsic motivator [14, 24]. This becomes less practical over time due to financial constraints, time, and data integrity [17]. Therefore, users must be continuously attracted and motivated to engage with the system to maintain accurate information. In contrast to gamified systems where users are awarded reputation points or badges for participation [18], the reward to engage users must be intrinsic [20, 21] and provide them with immediate insights or recommendations they can use outside of the system. To overcome external temporal factors causing data to evolve, these rewards need to create a feedback cycle that is beneficial for the user and the information [34]. Thus, the reward should be tightly coupled to user's implicit interactions, providing almost instantaneous motivation to contribute. These contributions create high-quality sources of information for the community.

5.2 Extending Drafty: DataBaits

Drafty requires a constant flow of visitors to visit, review data, and make unpaid contributions in the form of edits to maintain its evolving information. My previous research shows unpaid contributors can make accurate edits across data types requiring varying levels of domain expertise [34, 37]. While asking users to fix data matching their interests produces highly accurate edits, users only completed the request 9% of the time. Thus, Drafty requires a mechanism to motivate voluntary user contributions while not interrupting them during their natural information-seeking tasks. If a platform was as popular as Wikipedia or WikiData, it could rely on its popularity and real-world usage to attract visitors. While my theoretical framework could apply to Wikipedia and WikiData, it has been developed by and for researchers and developers who maintain smaller, less popular platforms to maintain evolving information.

I will extend Drafty with a new feature named DataBaits, to continuously attract and engage users by automatically providing random insights generated from Drafty's information after each edit. Drafty will tweet each DataBait with a link back to Drafty to continuously track users and understand if DataBaits can increase their engagement with Drafty. A DataBait is a statistic wrapped in sentence form. First, it takes a set of sample inputs from a tabular dataset. From Drafty's CS Professors dataset, it could use the columns "SubField" and "JoinYear" and the value "Human-Computer Interaction." This would produce a statistic about the change over time in the number of professors who specialize in Human-Computer Interaction: "Over the past 25 years, the total number of CS professors who specialized in Human-Computer Interaction increased 10 times." I have developed 15 different types of DataBaits covering ordinal and time-series data and historical events as inputs to ensure users see a random variety of insights.

5.3 Extending Remote Readability Tests: The Virtual Readability Lab

I have developed a suite of remote readability tests to study users' font preferences, reading speed, and reading comprehension while randomly changing the text's design [3, 32, 35, 36, 38]. Users are provided font recommendations after completing a remote readability test. These tests serve as the basis to develop the Virtual Readability Lab (VRL). How to personalize information design to augment human reading performance is the research question. One novel

contribution to date is that our tests normalize font size based on human perception [36].

Collecting enough information to accomplish this is daunting, as there are over 800,000 publicly available digital fonts. This information becomes out-of-date and less useful with the creation of new fonts, device types, and the ever-changing default fonts of software, browsers, and operating systems. While the digital reading population currently skews younger, this population will continue to grow and diversify in age. My preliminary work [36] shows that age is an essential factor when personalizing font choice in combination with other research [8, 9]. Maintaining and developing this set of evolving information can benefit typographers, designers, and readers better understand how to manipulate information design to augment reading performance.

My preliminary work recruits paid crowd workers to bootstrap this process. However, to build an extensive set of digital reading data, it will be necessary to attract visitors to voluntarily similar to LabInTheWild [26]. While the Virtual Readability Lab will allow users to compare their results with others, it will also personalize each readability test with randomized examples from a user's prior results. This way, every readability test will personalize the next, thus motivating users to take tests to contribute new information continuously.

6 CONCLUSION

My research focuses on developing and deploying asynchronous collaboration systems in the wild to maintain evolving information. These systems will provide random examples to the individual users that can help them immediately, thus motivating them to contribute new information continuously. These contributions create high-quality sources of information for the community. For example, Drafty's information helps power CS Open Rankings. Furthermore, Sketchy has empowered remote sketching activities during the COVID-19 pandemic. By attracting continuous contributions, the underlying information overcomes its temporal constraints to remain relevant and attract the attention of new users.

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